

S6 INSTRUMENTATION ENGINEERING***COURSE: INDUSTRIAL IoT****Time: 3 Hour**Max. Marks: 75****I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)***

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1	IoT are sensors that send real-time data to intermediate devices.	M1.03	R
2	 is the act of syncing data between two or more servers.	M1.02	U
3	is a network of nodes, designed to observe and to study physical parameters of the application.	M1.04	R
4	MQTT stands for	M2.01	R
5	 is a messaging protocol that is used for tracking devices in IoT.	M2.01	R
6	 split data into multiple collections and store it in multiple machines.	M3.01	R
7	An is a device which provides an entry point into enterprise or service provider core networks.	M3.04	U
8	A specification called can be used for storing high volume file into Arduino.	M4.01	R
9	Raspberry Pi is a complete mini.....	M4.02	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1.	List the characteristics of IoT.	M1.01	R
2.	List the fundamental components of IoT.	M1.02	R
3.	Differentiate between M2M and IoT.	M2.02	U
4.	Compare SDN and NFV.	M2.03	U
5.	Define the domain model specifications in IoT.	M3.02	U
6.	Illustrate the constituents of application development framework.	M3.04	U
7.	List the steps of component integration in IoT.	M3.03	R
8.	Compare Raspberry pi with Desktop Computer.	M4.02	U
9.	List the various components of Arduino Board.	M4.01	R
10.	List the features that can be included in an IoT based smart driving system.	M4.03	A

Part – C

Answer all questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
III	Describe the levels in IoT.	M1.05	U
	OR		
IV	Explain the communication protocols in IoT.	M1.04	U
V	With diagram explain IoT functional blocks.	M1.02	U
	OR		
VI	Describe about IoT protocols.	M1.03	U
VII	Explain Software Defined Networking architecture.	M2.03	U
	OR		
VIII	Explain the architecture of M2M.	M2.01	U
IX	Explain functional view specification in IoT.	M3.02	U
	OR		
X	Explain information model specification in IoT.	M3.02	U
XI	Describe Process specification in IoT.	M3.02	U
	OR		
XII	Explain service specification in IoT.	M3.02	U
XIII	Design Smart health care using IoT.	M4.03	A
	OR		
XIV	Design Smart water through IoT	M4.03	A

Question Wise Analysis

Course: **INDUSTRIAL INTERNET OF THINGS** – Question -1

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	M1.03	R	1	2
I.2	M1.02	U	1	2
I.3	M1.04	R	1	2
I.4	M2.01	R	1	2
I.5	M2.01	R	1	2
I.6	M3.01	R	1	2
I.7	M3.04	U	1	2
I.8	M4.01	R	1	2
I.9	M4.02	R	1	2
II.1	M1.01	R	3	8
II.2	M1.02	R	3	8
II.3	M2.02	U	3	8
II.4	M2.03	U	3	8
II.5	M3.02	U	3	8
II.6	M3.04	U	3	8
II.7	M3.03	R	3	8
II.8	M4.02	U	3	8
II.9	M4.01	R	3	8
II.10	M4.03	A	3	8
III	M1.05	U	7	16
IV	M1.04	U	7	16
V	M1.02	U	7	16
VI	M1.03	U	7	16
VII	M2.03	U	7	16
VIII	M2.01	U	7	16
IX	M3.02	U	7	16
X	M3.02	U	7	16
XI	M3.02	U	7	16
XII	M3.02	U	7	16
XIII	M4.03	A IoT	7	16
XIV	M4.03	A	7	16

Course : **INDUSTRIAL INTERNET OF THINGS –QP-1**

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Mark Distribution

Module	hr / Module	Marks/Module ($\sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	15	31 ± 6	3	3	2	6	4	28	9	37
2	14	28.5 ± 6	2	2	2	6	2	14	6	22
3	15	30.5 ± 6	2	2	3	9	4	28	9	39
4	16	33 ± 6	2	2	3	9	2	14	7	25
Total	60	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	19	15.5
Understanding	87	70.5
Applying	17	14
Analysing	0	0
Evaluating	0	0

Creating	0	0
Total	123	100

S6 INSTRUMENTATION ENGINEERING***COURSE: INDUSTRIAL IoT****Time: 3 Hour**Max. Marks: 75****I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)***

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1	is the most widely used protocol for data transmission in internet.	M1.03	R
2	refers to the strategy of analyzing large volumes of data.	M1.04	R
3	 is a low-energy version of Bluetooth protocol that reduces the power. consumption and plays an important role in connecting IoT devices.	M1.04	U
4	 is defined as the physical separation of the networking architecture of control plane from the data plane, and centralizes the network controller.	M2.03	R
5	 provides the infrastructure on which SDN can run.	M2.04	U
6	 model defines the attributes of the objects and relationships between objects.	M3.02	R
7	Representation of physical entity in digital world is known as	M3.04	R
8	Raspberry Pi is a low-cost mini-computer with the physical size of a	M4.02	R
9	Arduino devices arefor building digital devices that can sense and control objects in the physical and digital world.	M4.01	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1.	Define IoT.	M1.01	R
2.	List the basic building blocks of an IoT device.	M1.02	R
3.	List the features of application layer protocol.	M1.02	U
4.	List the advantages of SDN.	M2.03	R
5.	Define the process specification in IoT.	M3.02	U
6.	List the steps of IoT design methodology.	M3.01	U
7.	Define the service specification in IoT.	M3.02	R
8.	Compare Raspberry Pi with Arduino.	M4.01	U

9.	Write notes on the interfaces of Raspberry Pi.	M4.02	R
10.	List the features that can be included in an IoT based smart retail system.	M4.03	A

Part – C

Answer all questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
III	Describe about wireless sensor network.	M1.04	U
	OR		
IV	Explain the protocols in IoT.	M1.02	U
V	Explain IoT communication API's.	M1.03	U
	OR		
VI	Describe the features of cloud computing.	M1.04	U
VII	Explain Network Function Virtualization.	M2.04	U
	OR		
VIII	Explain the architecture of M2M.	M2.01	U
IX	Explain operational view specification in IoT.	M3.02	U
	OR		
X	Explain purpose and requirements specification in IoT.	M3.02	U
XI	Illustrate Arduino interfacing to LCD display.	M4.01	U
	OR		
XII	Design Smart driver assistance system using IoT.	M4.03	A
XIII	Design Smart warehouse monitoring using IoT.	M4.03	A
	OR		
XIV	Design Smart perishable tracking through IoT.	M4.03	A

Question Wise Analysis

Course: **INDUSTRIAL IoT** – Question -2

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	M1.03	R	1	2
I.2	M1.04	R	1	2
I.3	M1.04	U	1	2
I.4	M2.03	R	1	2
I.5	M2.04	U	1	2
I.6	M3.02	R	1	2
I.7	M3.04	R	1	2
I.8	M4.02	R	1	2
I.9	M4.01	R	1	2
II.1	M1.01	R	3	8
II.2	M1.02	R	3	8
II.3	M1.02	U	3	8
II.4	M2.03	R	3	8
II.5	M3.02	U	3	8
II.6	M3.01	U	3	8
II.7	M3.02	R	3	8
II.8	M4.01	U	3	8
II.9	M4.02	R	3	8
II.10	M4.03	A	3	8
III	M1.04	U	7	16
IV	M1.02	U	7	16
V	M1.03	U	7	16
VI	M1.04	U	7	16
VII	M2.04	U	7	16
VIII	M2.01	U	7	16
IX	M3.02	U	7	16
X	M3.02	U	7	16

XI	M4.01	U	7	16
XII	M4.03	A	7	16
XIII	M4.03	A	7	16
XIV	M4.03	A	7	16

Course: **INDUSTRIAL IoT – QP 2.**

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Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	15	31 ± 6	3	3	3	9	4	28	10	40
2	14	28.5 ± 6	2	2	1	3	2	14	5	19
3	15	30.5 ± 6	2	2	3	9	2	14	7	25
4	16	33 ± 6	2	2	3	9	4	28	9	39
Total	60	123	9	9	10	30	12	84	31	123

Blue Print
Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	22	18
Understanding	77	62.5
Applying	24	19.5
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100